

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Internet of things [ECE_4412]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare the distinct purposes and functionalities of Azure IoT Hub and SensorCloud. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) A customized surveillance system receives videos from multiple cameras distributed in the field via wireless channel and visualizes them on a shared LCD screen. Design a comprehensive protocol for validating the performance of the surveillance system. (5)
- Course Outcome: "CO4" / Cognitive Level: "Evaluate"
- 2) A smart home network connects a moderate number of devices like sensors, lights, and smart plugs within a limited, controlled area. Key needs include ease of setup, low power consumption, and moderate reliability. Evaluate the performance of different network topologies in this application context. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Justify the choice of specific network topology and modulation technique in LoRa. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Differentiate Eclipse Kura and openHAB. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Identify the tools within Amazon Web Services that are suitable for data storage and analysis in an IoT paradigm and evaluate the contextual application and suitability of each tool in terms of the specifications. (5)
- Course Outcome: "CO5" / Cognitive Level: "Create"
- 4) Demonstrate the application of Azure DevOps for designing an IoT application in which a wireless sensor network is deployed for environmental monitoring. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Design the basic layout of an IoT and cloud-inspired home automation system exclusively for HVAC, lighting, and basic home appliances control. (5)
- Course Outcome: "CO5" / Cognitive Level: "Create"

- 5) Illustrate two application contexts that can show the differences in edge computing and fog computing? (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) How does the separation of the control plane and the data plane, which is a foundational principle of Software-Defined Networking (SDN), enable the dynamic, programmable, and scalable software-defined cloud infrastructure? (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	15
2	2	CO 2	5
3	3	CO 3	5
4	4	CO 4	10
5	5	CO 5	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4412:Internet of things

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	3	15(30%)
4	Analysis	3	15(30%)
5	Create	1	5(10%)
6	Evaluate	2	10(20%)
Total		10	50 (100%)